

Household Food OS – UX Blueprint

1. Product Structure

Primary Navigation:

Home / Pantry / Planner / Community / Profile

Core User Jobs:

1. Decide what to cook today
2. Prevent ingredient expiry
3. Plan weekly meals
4. Achieve nutrition goals
5. Generate shopping decisions
6. Exchange surplus food locally

2. Information Architecture

HOME

- Today's Meal Recommendation
- Best / Quick / Lazy Options
- Expiry Alerts
- Nutrition Progress
- Shopping Recommendations
- Community Opportunities
- Impact Summary

PANTRY

- Inventory Overview
- Categories
- Expiry Timeline
- Scan Food
- Receipt Upload
- Order Import
- Manual Entry
- Inventory History

PLANNER

- Weekly Planner
- Daily Planner
- Meal Recommendations
- Recipe Detail
- Regenerate Plan
- Nutrition Tracking
- AI Conversation

COMMUNITY

- Map View
- Nearby Surplus Items
- Requests
- Offers
- Barter History
- Trust & Ratings

PROFILE

- Household Details
- Dietary Preferences
- Nutrition Goals
- Shopping Preferences
- Household Insights
- Settings

3. Core User Journey

ONBOARDING

Household Setup → Diet Setup → Goal Setup → Shopping Sources → Initial Kitchen Scan

DAILY LOOP

Open App → Review Today's Recommendation → View Recipe → Cook Meal → Feedback → Inventory Updated

WEEKLY LOOP

Review Weekly Plan → Accept/Modify Meals → Generate Shopping Suggestions → Purchase Food → Inventory Updated

EXPIRY LOOP

Ingredient Near Expiry → Alert Generated → Suggested Usage Meal → Consume Ingredient → Waste Prevented

COMMUNITY LOOP

Identify Surplus → Create Listing → Nearby Match → Exchange Completed → Trust Score Updated

4. Detailed Task Flow: Pantry Creation

User taps Scan Food

- Choose Scan Method
- Fridge Photo / Pantry Photo / Receipt / Grocery Import / Manual Entry
- AI Detects Items
- User Confirms Items
- AI Categorizes Inventory
- AI Assigns Expiry Estimates
- Pantry Updated
- Home Dashboard Refreshed

5. Detailed Task Flow: Daily Meal Decision

User opens Home

- System evaluates inventory
- System evaluates expiry risk
- System evaluates dietary goals
- System evaluates household preferences
- AI generates:
 - Best Option
 - Quick Option
 - Lazy Option
- User selects meal

- Recipe Detail
- Cooking
- User Feedback
- Inventory Consumption Logged
- Learning Model Updated

6. Detailed Task Flow: Weekly Meal Planning

User opens Planner

- Select Week
- AI evaluates pantry inventory
- AI evaluates nutrition targets
- AI evaluates household preferences
- AI creates weekly plan
- User edits meals if required
- Missing ingredients calculated
- Shopping list generated
- Weekly plan saved

7. Detailed Task Flow: Shopping Assistant

Inventory Forecast

- Predicted Shortages
- Weekly Plan Requirements
- Existing Stock Validation
- Purchase Recommendation
- Cart Generation
- Redirect to Grocery Partner
- Purchase Complete
- Inventory Updated

8. Detailed Task Flow: Community Exchange

User identifies surplus ingredient

- Create Offer
- AI recommends nearby recipients
- Listing visible on map
- Request accepted
- Exchange completed
- Rating submitted
- Community history updated

9. AI Interaction Model

SYSTEM MEMORY

- Inventory State
- Consumption Patterns
- Shopping Behaviour
- Preferred Meals
- Rejected Meals
- Nutrition Progress
- Community Behaviour

AI INPUTS

- Inventory
- Expiry Data
- Household Profile
- Nutrition Goals
- Shopping History
- Community Availability

AI OUTPUTS

- Meal Recommendations
- Shopping Suggestions
- Expiry Alerts
- Nutrition Guidance
- Community Matches
- Impact Metrics

10. AI Behaviour Framework

HOME

AI prioritizes highest value daily decision.

PANTRY

AI maintains inventory confidence score.

PLANNER

AI balances:

- expiry reduction
- nutrition
- convenience
- preference

SHOPPING

AI minimizes over-purchasing.

COMMUNITY

AI matches supply and demand.

PROFILE

AI continuously updates household model.

11. Screen-to-System Relationships

HOME ← Pantry + Planner + Community + Profile

PANTRY → feeds Planner

PLANNER → feeds Shopping

SHOPPING → feeds Pantry

COMMUNITY → feeds Pantry

PROFILE → influences all recommendations

12. Edge Cases

Unknown ingredient detection
Duplicate ingredients
Multiple users editing inventory
Expired ingredient confirmation
Out-of-stock recipe ingredients
Low confidence inventory state
No community matches available
Conflicting household goals

13. Future System Architecture

PHASE 1

- Inventory Intelligence
- Expiry Intelligence
- Meal Suggestions

PHASE 2

- Weekly Planning
- Nutrition Engine
- Shopping Assistant

PHASE 3

- Community Exchange
- Household Food Graph
- Impact Dashboard